Research Proposal Presentation

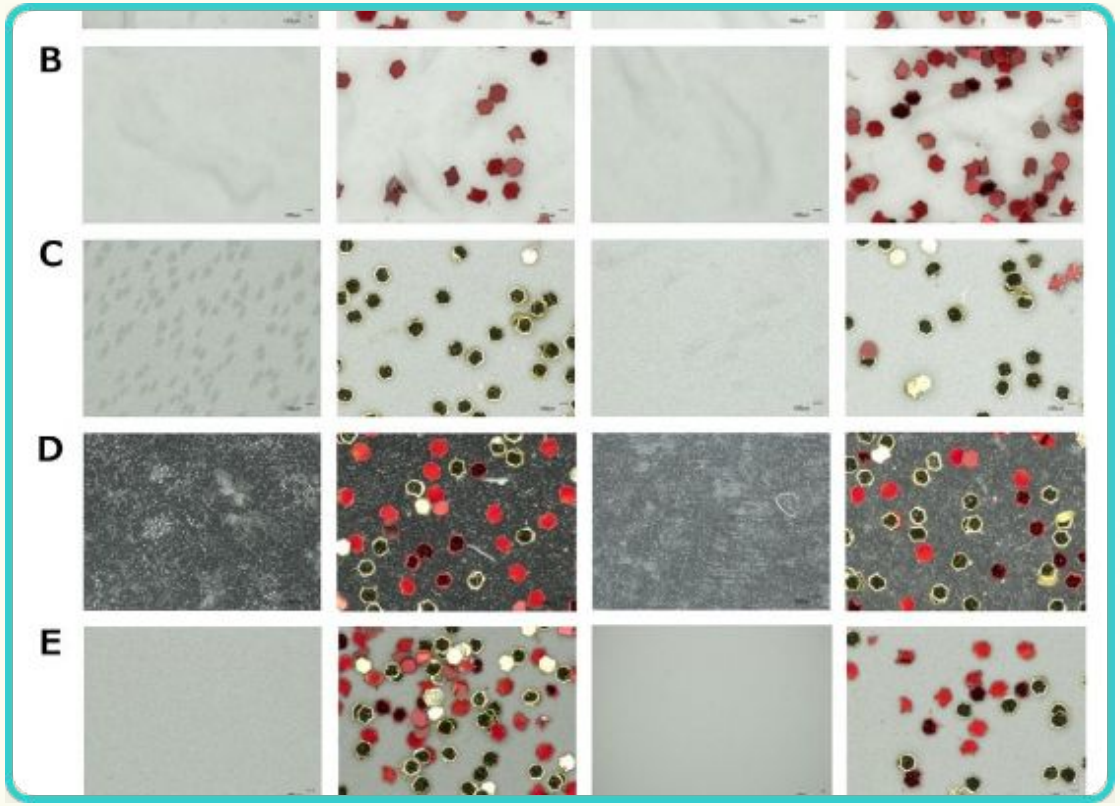
Dietary Exposure Risk: Predicting Blood Microplastic Loads

A Data-Driven ML Approach to Bioaccumulation Prediction

Expert Academic Researcher & Data Scientist Perspective

Background & Motivation

- Microplastics (MPs) are ubiquitous environmental contaminants detected in human lungs, liver, and placenta.
- Recent studies (2022–2024) confirm MP presence in the human bloodstream.
- Urgency: Need for individual risk profiling based on daily dietary and lifestyle factors.



Problem Statement

- **Quantification Gap:** High variability in MP detection results; lack of direct linkage to dietary source metrics.
- **Health Risk:** Chronic bioaccumulation is linked to inflammation and oxidative stress, yet exposure levels are unmonitored.
- **Accessibility:** Current blood testing for MPs is invasive, expensive, and unavailable to the general public.

Research Objectives

ML Task

Build a robust multi-class classifier to predict biological MP risk levels.

Key Pathways

Map high-risk "entry routes" like plastic-packaged foods and seafood.

Scalability

Create a non-invasive screening tool based on dietary intake metadata.

Research Hypotheses

Hypothesis 1

Individuals consuming >4 items of plastic-packaged processed food daily show a 70% higher probability of "High Risk" classification.

Hypothesis 2

Fresh, plant-based diets serve as a predictive indicator for "Negligible" or "Low" MP load, regardless of urban proximity.

Literature Context

"For the first time, we have shown that microplastics are in human blood. This study suggests that we are ingesting plastic daily through various environmental vectors."

— Leslie et al. (2022), Environment International

Gap: Existing studies focus on environmental detection; this research moves toward behavioral prediction.

Dataset Description

Data Category	Variable Count	Key Features
Food Consumption	12+ Features	Salt, Honey, Seafood, Bottled Water, Canned Goods
Packaging Type	Categorical	PET, LDPE, HDPE, Non-Plastic (Glass/Paper)
Demographics	4 Features	Age, Urban/Rural Setting, Water Source
Target Variable	**3 Classes**	**Exposure Risk (Negligible, Low, High)**

Feature Engineering Strategy

Plastic Load Index (PLI): An aggregated score calculated as:

$$PLI = \sum_{i=1}^{n} (Consumption_i \cdot Packaging_Weight_i)$$

Encoding Packaging types using Ordinal Encoding (Risk-based).

Methodology Pipeline

Pre-processing

Outlier removal & Missing value imputation

Sampling

SMOTE for class balancing

Modeling

Ensemble Tree Training (XGBoost)

Tuning

Optuna Hyperparameter Search

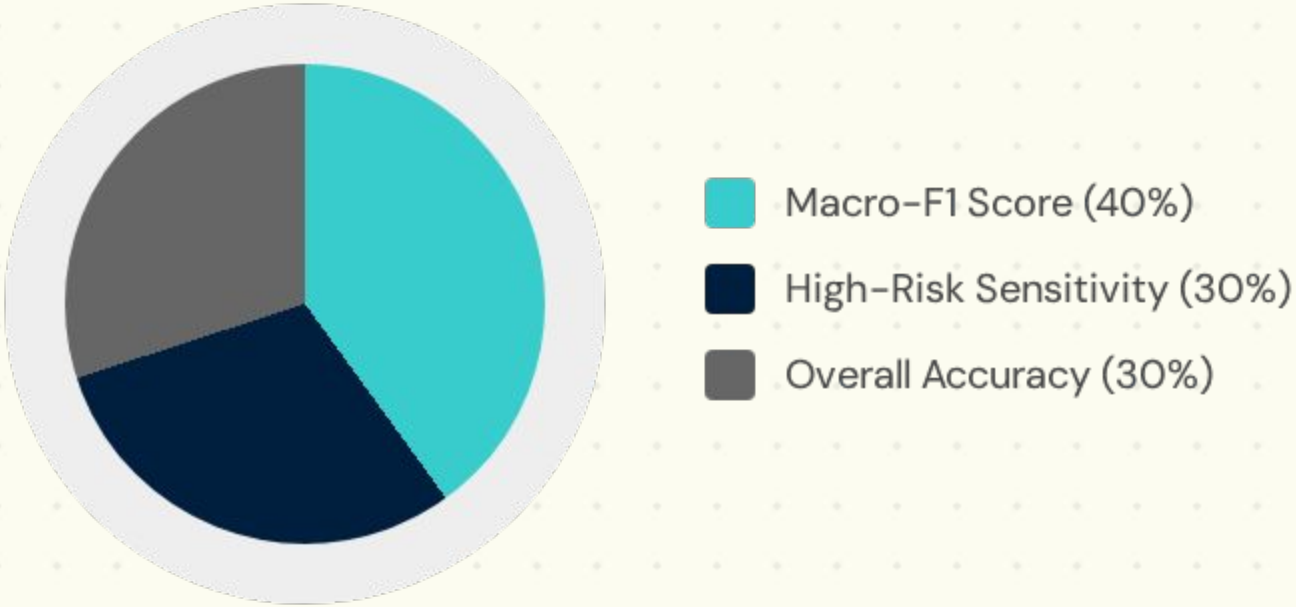
Validation

K-Fold Cross Validation

Model Choice & Performance

XGBoost is selected for its superior handling of non-linear dietary interactions and class-weighted optimization.

Metric Priority Matrix



Priority is given to **Sensitivity for High Risk** to avoid false negatives in clinical profiling.

Anticipated Impact Trends

Definition	• Minimal processing includes the removal of inedible or unwanted parts of a food source and very basic preservation like drying. • Nothing should be added to the original food in this category at the point of sale.	• This category was broadly created to bridge the gap between ready-made and home cooked foods, comprising the ingredients like salt, herbs, sugar, vinegar and oil that are typically added during cooking but do not constitute the bulk of the meal. • These foods are rarely eaten by themselves and unlike the other three categories are not grouped by the way in which they are made, but rather by the way in which they are used.	• Foods from group one that have been processed and/or combined with foods from group two, or group one or two foods that have been further processed into a final product that is ready to cook or eat. • There are generally fewer processing steps and ingredients in processed compared to ultra-processed foods, although there can be overlap in the processes and ingredients used, and grey areas in dividing foods between the two.	• Foods with many ingredients in them and multiple processing steps that could not be recreated in a conventional kitchen. • These foods are often (though not necessarily) high in salt, sugar and fat, low in fibre and **calorie dense**. • These foods are generally convenient and quick to prepare, cheap, tasty, and often have plastic packaging.
Examples	• Fruits, vegetables, beans, mushrooms,	• Salt, peper, herbs, cooking oil, sugar, vanilla extract,	• Sourdough bread, canned chickpeas, tofu, tempeh,	• Energy drinks, hot dogs, chocolate bars, shaped salty

Key Trend: Anticipate a log-linear relationship between bottled water consumption frequency and blood MP load across the "Urban" demographic segment.

Novelty & Contributions

1st

90%

Zero

Bridges the gap between data science and molecular environmental toxicology.

Limitations & Ethics

Limitations

- Recall bias in self-reported diet
- Excludes inhalation pathways
- Metabolic variability not modeled

Ethics

- Data privacy (GDPR compliance)
- Non-diagnostic disclaimer
- Algorithmic bias in food access

Conclusion & Future Work

Next Steps

- App integration for consumer risk tracking.
- Longitudinal study on organ health correlation.
- Collaboration with environmental policy makers.

Questions?

Thank you for your attention.

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